TEACHER COPY — QUESTIONS WITH KEY ANSWERS

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IT 10 — 1st Quarter Examination

Question Bank (77 items) with Answer Key · Alan L. Vergara

Each attempt draws 50 items: the **12 required items (Part I picture ID + Part IV lab tasks) always appear**; 38 are drawn unseen-first from the rest.
Identification items accept the variants shown after "Key:". Part I items display a photograph of the keyed component.
Part IV items open the embedded Virtual Arduino Lab and are graded against the rubric shown.

Domain A — Arduino Programming Fundamentals (25 items)

1. Which one is the fastest Led blinking?

a) delay(500);
b) delay(1000);
c) delay(1500);

Key: **delay(500);**

2. Which one of the following is a user defined function?

a) loop();
b) calculate();
c) setup();

Key: **calculate();**

3. Which function runs only one time?

a) loop();
b) calculate();
c) setup();

Key: **setup();**

4. To pause a system for 2 seconds in Arduino, we use which line of code?

a) delay(20);
b) delay(200);
c) delay(2000);

Key: **delay(2000);**

5. How many analog pins are on an Arduino board?

a) 5
b) 6
c) 7

Key: **6**

6. How many digital pins are on an Arduino board?

a) 10
b) 12
c) 14

Key: **14**

7. All of these are input devices EXCEPT

a) Temp Sensor
b) LED
c) Ultrasonic Sensor

Key: **LED**

8. What is the result of the following code: print(7%3);

a) 0
b) 1
c) 2

Key: **1**

9. The following line of code means pinMode(12, OUTPUT);

a) Set the device in pin 12 as output.
b) Set the device in pin 12 as input.
c) Set the device in pin 12 set to Off state.

Key: **Set the device in pin 12 as output.**

10. To control the current through a circuit we use

a) Wire
b) Sensor
c) Resistor

Key: **Resistor**

11. Code writing and uploading into the Arduino board is done using which tool?

a) IDE
b) PC
c) SOME

Key: **IDE**

12. The Arduino board is also called a

a) Timer
b) Oscillator
c) Microcontroller

Key: **Microcontroller**

13. To ground the circuit, which pin is used in Arduino?

a) RESET
b) GND
c) PIN

Key: **GND**

14. Arduino IDE consists of 2 functions. What are they?

a) build() & loop()
b) setup() & build()
c) setup() & loop()

Key: **setup() & loop()**

15. digitalWrite(13, LOW); turns the LED on pin 13

a) ON
b) OFF
c) DIM

Key: **OFF**

56. Arduino Uno digital pin 13 has a built-in _____.

a) LED
b) button
c) motor

Key: **LED**

57. Which of these is a function used to send text over the serial port?

a) Serial.print()
b) delay()
c) pinMode()

Key: **Serial.print()**

58. analogRead() returns values in which range?

a) 0 to 1023
b) 0 to 255
c) 0 to 5

Key: **0 to 1023**

59. digitalWrite() can set a digital pin to

a) HIGH or LOW
b) 0 to 1023
c) positive or negative

Key: **HIGH or LOW**

60. A baud rate commonly used with Serial.begin() is

a) 9600
b) 1200
c) 96000

Key: **9600**

61. The loop() function in Arduino

a) runs repeatedly
b) runs once
c) never runs

Key: **runs repeatedly**

62. With pinMode(pin, INPUT_PULLUP), a pushbutton reads _____ when pressed.

a) LOW
b) HIGH
c) 5 volts

Key: **LOW**

63. Which unit measures electrical resistance?

a) ohm
b) volt
c) ampere

Key: **ohm**

64. The Arduino Uno receives power over USB at approximately how many volts?

a) 5
b) 12
c) 9

Key: **5**

65. Which Arduino function configures a pin as an output?

a) pinMode()
b) analogRead()
c) map()

Key: **pinMode()**

Domain B — Resistor Color Code (10 items)

16. Decode the value: Brown, Black, Red, Gold.

a) 1 kΩ ±5%
b) 100 Ω ±5%
c) 10 kΩ ±5%

Key: **1 kΩ ±5%**

17. Decode the value: Yellow, Violet, Yellow, Gold.

a) 470 kΩ ±5%
b) 47 kΩ ±5%
c) 4.7 MΩ ±5%

Key: **470 kΩ ±5%**

18. Decode the value: Brown, Black, Green, Gold.

a) 1 MΩ ±5%
b) 100 kΩ ±5%
c) 10 MΩ ±5%

Key: **1 MΩ ±5%**

19. Decode the value: Blue, Grey, Black, Gold.

a) 68 Ω ±5%
b) 680 Ω ±5%
c) 6.8 kΩ ±5%

Key: **68 Ω ±5%**

20. Decode the value: Orange, White, Orange, Gold.

a) 39 kΩ ±5%
b) 390 kΩ ±5%
c) 3.9 kΩ ±5%

Key: **39 kΩ ±5%**

21. Decode the value: Grey, Red, Gold, Gold.

a) 8.2 Ω ±5%
b) 82 Ω ±5%
c) 820 Ω ±5%

Key: **8.2 Ω ±5%**

22. Decode the value: Yellow, Violet, Green, Silver.

a) 4.7 MΩ ±10%
b) 47 MΩ ±10%
c) 4.7 kΩ ±10%

Key: **4.7 MΩ ±10%**

23. Decode the value: Yellow, Violet, Black, Gold.

a) 47 Ω ±5%
b) 470 Ω ±5%
c) 4.7 kΩ ±5%

Key: **47 Ω ±5%**

24. Decode the value: Orange, Black, Red, Silver.

a) 3 kΩ ±10%
b) 30 kΩ ±10%
c) 300 Ω ±10%

Key: **3 kΩ ±10%**

25. Decode the value: Green, Blue, Yellow, Silver.

a) 560 kΩ ±10%
b) 56 kΩ ±10%
c) 5.6 MΩ ±10%

Key: **560 kΩ ±10%**

Domain C — Resistance Value Computation (Ohm's Law) (10 items)

26. A 12 V source pushes 60 mA through a resistor. What is its resistance?

a) 200 Ω
b) 2 kΩ
c) 20 Ω

Key: **200 Ω**

27. What resistance draws 10 mA from a 5 V source?

a) 500 Ω
b) 5 kΩ
c) 50 Ω

Key: **500 Ω**

28. What resistance draws 33 mA from a 3.3 V source?

a) 10 Ω
b) 100 Ω
c) 1 kΩ

Key: **100 Ω**

29. What resistance draws 80 mA from a 24 V source?

a) 30 Ω
b) 300 Ω
c) 3 kΩ

Key: **300 Ω**

30. What resistance draws 30 mA from a 1.5 V source?

a) 5 Ω
b) 50 Ω
c) 500 Ω

Key: **50 Ω**

31. What resistance draws 40 mA from a 6 V source?

a) 15 Ω
b) 150 Ω
c) 1.5 kΩ

Key: **150 Ω**

32. What resistance draws 55 mA from a 110 V source?

a) 200 Ω
b) 2 kΩ
c) 20 kΩ

Key: **2 kΩ**

33. What resistance draws 2 mA from a 3 V source?

a) 150 Ω
b) 1.5 kΩ
c) 15 kΩ

Key: **1.5 kΩ**

34. What resistance draws 50 mA from a 20 V source?

a) 40 Ω
b) 400 Ω
c) 4 kΩ

Key: **400 Ω**

35. What resistance draws 64 mA from a 16 V source?

a) 25 Ω
b) 250 Ω
c) 2.5 kΩ

Key: **250 Ω**

Domain D — Arduino Family & SIK Components — Identification (20 items)

36. It is an open-source electronics platform based on easy-to-use hardware and software.

Key: **arduino** *(typed answer)*

37. The tiny computer on the Arduino board is called a _____.

Key: **microcontroller** *(typed answer)*

38. A solderless board used to build temporary prototypes without soldering.

Key: **breadboard** *(typed answer)*

39. Wires with a pin on both ends used to connect components on a breadboard.

Key: **jumper wires | jumper wire | jumper** *(typed answer)*

40. A three-terminal resistor with a rotating shaft used to adjust resistance.

Key: **potentiometer | trimmer | trimpot** *(typed answer)*

41. This component allows electricity to flow in only one direction.

Key: **diode | signal diode | rectifier diode** *(typed answer)*

42. The LED's longer leg is called the _____.

Key: **anode** *(typed answer)*

43. The LED's shorter leg is called the _____.

Key: **cathode** *(typed answer)*

44. How many ohms is a 10 kΩ resistor? Write digits only.

Key: **10000 | 10,000** *(typed answer)*

45. The instrument used to measure electrical resistance is called a _____.

Key: **multimeter | ohmmeter | multi meter** *(typed answer)*

46. Color code: a resistor with bands Red-Red-Brown-Gold is how many ohms? (digits only)

Key: **220** *(typed answer)*

47. The SIK sensor that measures nearby object distance using sound waves is the _____ sensor.

Key: **ultrasonic | sonar** *(typed answer)*

48. The SIK temperature sensor (TMP36) outputs a voltage proportional to _____.

Key: **temperature | heat** *(typed answer)*

49. A light-dependent resistor changes its resistance based on _____.

Key: **light | brightness | light intensity** *(typed answer)*

50. This SIK component converts electrical energy into continuous mechanical motion.

Key: **motor | dc motor | direct current motor** *(typed answer)*

51. A programmable light that can glow in many colors using three channels is called a _____ LED.

Key: **rgb** *(typed answer)*

52. The pin used to reset the Arduino board is the _____ pin.

Key: **reset** *(typed answer)*

53. The pin that provides 0 volts reference is the _____ pin.

Key: **gnd | ground** *(typed answer)*

54. The pin that provides regulated +5V on an Arduino Uno is the _____ pin.

Key: **5v | 5v power | power** *(typed answer)*

55. A semiconductor component that amplifies or switches electronic signals is a _____.

Key: **transistor** *(typed answer)*

Part I — Visual Component Identification (10 items) *required · picture shows the keyed component*

66. Identify the component shown in the picture.

Key: **arduino uno | arduino | arduino board | uno board | uno** *(typed answer)*

67. Identify the component shown in the picture.

Key: **ldr | light dependent resistor | photoresistor | light-dependent resistor** *(typed answer)*

68. Identify the component shown in the picture.

Key: **lcd | lcd display | liquid crystal display | 16x2 lcd | character lcd** *(typed answer)*

69. Identify the component shown in the picture.

Key: **buzzer | piezo buzzer | piezo | sound module** *(typed answer)*

70. Identify the component shown in the picture.

Key: **breadboard | solderless breadboard** *(typed answer)*

71. Identify the component shown in the picture.

Key: **led | light emitting diode | light-emitting diode | leds** *(typed answer)*

72. Identify the component shown in the picture.

Key: **dc motor | motor | direct current motor** *(typed answer)*

73. Identify the component shown in the picture.

Key: **resistor | resistors** *(typed answer)*

74. Identify the component shown in the picture.

Key: **pushbutton | push button | push button switch | button | tactile switch** *(typed answer)*

75. Identify the component shown in the picture.

Key: **servo | servo motor | sg90 | servo motor sg90** *(typed answer)*

Part IV — Problem Solving (Virtual Arduino Lab) (2 items) *required · rubric-scored*

76. Part IV — Problem Solving (5 pts). Write a complete Arduino sketch that will turn an LED on and off using a push button. Build and test your circuit in the Virtual Arduino Lab below, then paste your final code here.

Key: **Rubric: A working sketch that sets pinMode() for the pushbutton (INPUT_PULLUP or an external pull-down) and the LED pin as OUTPUT, reads the button inside loop(), and uses digitalWrite() to turn the LED ON and OFF. | Keywords: pinmode; digitalwrite; loop; button=pushbutton, switch; led** *[Virtual Arduino Lab — manual, rubric-scored]*

77. Part IV — Problem Solving (10 pts). Build the Arduino circuit that turns an LED on and off using a push button. Construct it yourself on the simulation canvas in the Virtual Arduino Lab below, then describe your connections here — which Arduino pins you used and where each component goes.

Key: **Rubric: A correct circuit: Arduino wired to a breadboard, LED with a series current-limiting resistor (220 Ω to 1 kΩ), pushbutton between a digital pin and GND (INPUT_PULLUP or pull-down), LED anode to a digital pin and cathode to GND, all grounds common. | Keywords: breadboard; resistor; button=pushbutton, switch; gnd=ground, gnd; pin=pin, pinmode, digital; led** *[Virtual Arduino Lab — manual, rubric-scored]*